

MA1521 L1 Week 03 Wed 20260826

For each of the following vector functions of time, calculate the velocity, speed, unit tangent vector (in the direction of velocity), and acceleration.

a) $e^t \mathbf{i} + e^{-t} \mathbf{j}$ b) $t^2 \mathbf{i} + t^3 \mathbf{j}$ c) $(1 - 2t^2) \mathbf{i} + t^2 \mathbf{j} + (-2 + 2t^2) \mathbf{k}$

Prove the rule for differentiating the scalar product of two plane vector functions:

$$\frac{d}{dt} \mathbf{r} \cdot \mathbf{s} = \frac{d\mathbf{r}}{dt} \cdot \mathbf{s} + \mathbf{r} \cdot \frac{d\mathbf{s}}{dt} ,$$

by calculating with components, letting $\mathbf{r} = x_1 \mathbf{i} + y_1 \mathbf{j}$ and $\mathbf{s} = x_2 \mathbf{i} + y_2 \mathbf{j}$.

Suppose a point P moves on the surface of a sphere with center at the origin; let

$$OP = \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}.$$

Show that the velocity vector $\mathbf{v}$ is always perpendicular to $\mathbf{r}$

Prove the converse: if $\mathbf{r}$ and $\mathbf{v}$ are perpendicular, then the motion of P is on the surface of a sphere.

a) Prove the formula $\frac{d}{dt} u(t)\mathbf{r}(t) = \frac{du}{dt} \mathbf{r}(t) + u(t) \frac{d\mathbf{r}}{dt}.$

(You may assume the vectors are in the plane; calculate with the components.)

b) Let $\mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j}$, the exponential spiral. Use part (a) to find the speed of this motion.